Question 3 | International Pumpkin Pasties

Economic historians in the wizarding world remember a time before the Ministry had begun meddling with the domestic market. In those earlier days, trade barriers had been lifted for exports of pumpkin pasties to international magical markets with a world price of 12 galleons. At the time, the Ministry of Magic used this opportunity to pad its coffers without facing the political backlash associated with a domestic tax, instead imposing an import tariff of 2 galleons. Use a graph to illustrate what happened to the market after the tariff was imposed. Remember the supply and demand curves:

$$P = 17 - \frac{1}{6}Q_D$$

$$P = 2 + \frac{2}{3}Q_S$$

$17 - \frac{1}{6}Q_D = 2 + \frac{2}{3}Q_S$

$15 = \left(\frac{1}{6} + \frac{4}{6}\right)Q = \frac{5}{6}Q$

$Q = \frac{15}{5} \cdot 6 = 18 = Q^*$

$P = 2 + \frac{2}{3} \cdot 18 = 2 + 12$

$P^* = 14$

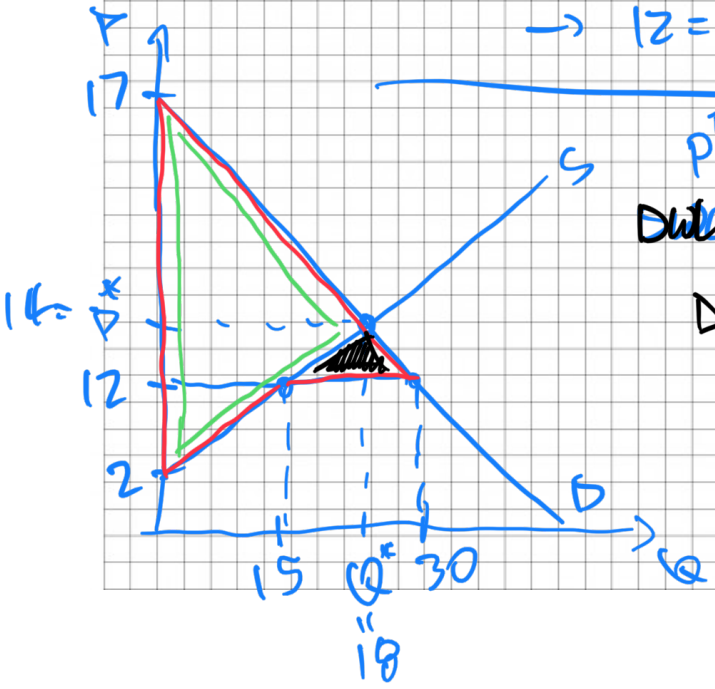
@ 12 → $12 = 17 - \frac{1}{6}Q_D \Rightarrow \frac{1}{6}Q_D = 5 \Rightarrow Q_D = 30$

→ $12 = 2 + \frac{2}{3}Q_S \Rightarrow \frac{2}{3}Q_S = 10 \Rightarrow Q_S = 15$

$P^T = 7 + P_G \Rightarrow P^T = 14$

$DWL = h \cdot (Q_D - Q_S) \cdot \frac{1}{2} = 2(30-15) \cdot \frac{1}{2}$

$DWL = 15$



What is the area of the deadweight loss from the tariff? 15

After the tariff, the price domestically went back to 14, with no imports, and DWL in black.